

Problem

The initial value of $f(t)$ with transform

$$F(s) = \frac{s+1}{(s+2)(s+3)}$$

is:

- (a) nonexistent
- (b) ∞
- (c) 0
- (d) 1
- (e) $\frac{1}{6}$

Step-by-step solution

Step 1 of 1

$$F(S) = \frac{S+1}{(S+2)(S+3)} = \frac{A}{S+2} + \frac{B}{S+3}$$

$$S+1 = (A+B)S + 3A + 2B$$

$$\Rightarrow A = 1 - B, \quad 3A + 2B = 1$$

$$3A - 2A = -1$$

$$\Rightarrow A = -1$$

$$B = 1 - (-1) = 2$$

$$F(S) = \frac{-1}{S+2} + \frac{2}{S+3}$$

$$f(t) = -(e^{-2t}) + 2e^{-3t}$$

$$\begin{aligned} \text{Initial value } f(0) &= -e^{-0} + 2e^{-0} \\ &= -1 + 2 = 1 \end{aligned}$$